

# Nayoung Ku

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## EDUCATION

### Handong Global University

*B.Sc. in Life Science and Applied Artificial Intelligence (cum laude, GPA: 4/4.5)*

North Gyeongsang, South Korea

Feb. 2023 – Feb. 2026

### National Institute for Lifelong Education

*B.L.A. in Psychology (GPA: 4/4.3)*

Seoul, South Korea

Feb. 2021 – Feb. 2023

## HONORS & AWARDS

### 3rd Place, Daewoong Pharmaceutical President's Award (out of 990 teams)

2024

*JUMP AI Drug Discovery Competition – Organized by KPBMA & Sponsored by the Ministry of Health and Welfare*

### Big Data Seagull Scholarship

Spring 2024, Fall 2024

*Merit-based scholarship, certified by HGU School of Applied Artificial Intelligence*

## TECHNICAL SKILLS

**Languages:** Python, R, SQL, Bash

**ML/DL:** PyTorch, scikit-learn, Weights & Biases, MLflow, Optuna, HuggingFace

**Bioinformatics:** Scanpy, scRNA-seq Analysis, Bulk RNA-seq, Histopathology & Microscopy Image Analysis

**Infrastructure:** Git, GitHub, Slurm, LSF, MIG-partitioned GPU clusters, Linux (Ubuntu, ARM)

## RESEARCH EXPERIENCE

### Undergraduate Researcher

Aug. 2024 – Present

*Handong Global University – Advisor: Prof. Jaeyoung Chun*

*North Gyeongsang, South Korea*

- Designing a gene regulatory network-aware **genetic perturbation (CRISPRi)** effect prediction model to capture **transcriptomic shifts** for **multiple cell-line generalization**
- Standardized heterogeneous **Perturb-seq** datasets and built a reproducible model training/evaluation pipeline
- Developed a diffusion-based probabilistic demultiplexing pipeline for hashed scRNA-seq data that learns the underlying signal distribution without hard-threshold cutoffs
- Constructed a high-quality annotated training dataset for **circulating tumor cell (CTC)** detection in unstained blood **microscopy images** and conducted a systematic literature review on CTC morphology
- Initiated and led an **IC50 predictive modeling** from chemical structure (SMILES), generated features, and designed feature selection methods for ML regression models
- Achieved **top 3 finalist** status at a national AI drug discovery challenge (see Honors & Awards) and **presented the poster at AI Pharma Korea Conference 2024**

### Visiting Undergraduate Researcher

Jun. 2025 – Jul. 2025

*University of Nevada, Las Vegas – Advisor: Prof. Mingon Kang (DataX Lab)*

*Nevada, United States*

- Developed a lightweight **patch tiling and filtering algorithm** for **whole slide images (WSIs)** from the **TCIA CMB-LCA**, reducing raw data volume by **80%** (128GB → 25GB), significantly lowering computational overhead
- Built a hybrid deep learning framework integrating **Vision Transformer** embeddings with an **LSTM**-based classifier, achieving **AUC 0.99** on large-scale, imbalanced tumor classification across WSIs

### Undergraduate Researcher

Jan. 2025 – Apr. 2025

*Seoul National University – Advisor: Prof. Dr. Howard Lee (CCADD)*

*Seoul, South Korea*

- Contributed to a **NLP research project commissioned by Korea's Ministry of Food and Drug Safety (MFDS)**, building a **Korean-English parallel sentence corpus** of pharmaceutical regulatory documents
- Standardized data collection, annotation, and management protocols across a multi-source document corpus, establishing a reusable curation pipeline for the research team, and designed a sentence alignment algorithm for automated parallel corpus construction from heterogeneous document sources

### Undergraduate Research Intern

Mar. 2024 – Nov. 2024

*Handong Global University – Advisor: Prof. Taejin Ahn (BioData Lab)*

*North Gyeongsang, South Korea*

- Analyzed clinical **Tumor-Educated Platelet (TEP) RNA-seq** datasets to identify **ovarian and colorectal cancer biomarkers**, applying feature selection and hierarchical clustering to isolate signal from high data sparsity
- Developed a **transformer-based preprocessing framework** encoding raw RNA sequences of TEPs directly into embeddings without traditional count-matrix construction and achieved **AUC 0.71** in ovarian cancer classification

## PRESENTATIONS

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### *Conference & Capstone Presentations*

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|------------------------------------------------------------------------------------------------|------------------------------|
| <b>No Cell Left Behind: Recovering Discarded Cells via Self-Supervised Diffusion Denoising</b> | Dec. 2025                    |
| <i>Handong Global University Bio-AI Capstone Design Fair</i>                                   | <i>Capstone Presentation</i> |
| <b>Diagnosing Cancer with Genomic Context</b>                                                  | Nov. 2024                    |
| <i>Handong Global University Capstone Design in Life Science Fair</i>                          | <i>Capstone Presentation</i> |
| <b>IRAK4 IC50 Inhibition Prediction</b>                                                        | Oct. 2024                    |
| <i>AI Pharma Korea Conference 2024</i>                                                         | <i>Poster Presentation</i>   |

## TEACHING EXPERIENCE

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|--------------------------------------------------------------------------------------|------------------------------|
| <b>Introduction to AI and Machine Learning, <i>Teaching Assistant</i></b>            | Aug. 2025                    |
| <i>Addis Ababa Science and Technology University – Instructor: Prof. Dai Yon Cho</i> | <i>Addis Ababa, Ethiopia</i> |
| <b>Big Data Modeling and Platforms, <i>Teaching Assistant</i></b>                    | Mar. 2025 – Jun. 2025        |
| <i>Handong Global University – Instructor: Prof. Jaeyoung Chun</i>                   | <i>Pohang, South Korea</i>   |
| <b>AI Application Programming, <i>Teaching Assistant</i></b>                         | Mar 2025. – Jun. 2025        |
| <i>Handong Global University – Instructor: Prof. Sangsan Lee</i>                     | <i>Pohang, South Korea</i>   |

## COMMUNITY INVOLVEMENT

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|---------------------------------------------------------------------------------------------|-----------------------------|
| <b>R Programming Camp, <i>Teaching Assistant</i></b>                                        | Jan 2026                    |
| <i>Mentored high school students in data analysis and visualization using R</i>             | <i>Pohang, South Korea</i>  |
| <b>Handong Office of International Affairs, <i>Exchange Student Mentor</i></b>              | Aug – Dec 2024              |
| <i>Assisted exchange students with adaptation to Korean culture and residential college</i> | <i>Pohang, South Korea</i>  |
| <b>Incheon Beauty and Art High School Python Camp, <i>Teaching Assistant</i></b>            | Aug 2023                    |
| <i>Supported Python programming instruction for elementary to high school students</i>      | <i>Incheon, South Korea</i> |